

assignment_rotation1

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:46pm

Quiz Instructions

no tutorial



Question 1 12 pts

When a rigid body rotates about a fixed axis, all the points in the body have the same

☐

angular acceleration.

☐

tangential acceleration.

☐

tangential speed.

☐

linear displacement.

☐

centripetal acceleration.



Question 2 12 pts

How long does it take for a rotating object to speed up from 15.0 rad/s to 33.3 rad/s if it has a uniform angular acceleration of 3.45 rad/s^2 ?

☐

5.30 s

☐

4.35 s

☐

63.1 s

☐

9.57 s

☐

10.6 s



Question 3 12 pts

A child is riding a merry-go-round that has an instantaneous angular speed of 1.25 rad/s and an angular acceleration of 0.745 rad/s^2 . The child is standing 4.65 m from the center of the merry-go-round. What is

the magnitude of the linear acceleration of the child?

☐

7.27 m/s²

☐

2.58 m/s²

☐

3.46 m/s²

☐

4.10 m/s²

☐

8.05 m/s²

☐

Question 4 12 pts

A wheel accelerates with a constant angular acceleration of 4.5 rad/s² from an initial angular speed of 1.0 rad/s. (a) Through what angle does the wheel turn in the first 2.0 s, and (b) what is its angular speed at that time?

☐

(a) 11 rad/s (b) 20 rad/s

☐

(a) 10rad (b) 20 rad/s

☐

(a) 11 rad (b) 10 rad/s

☐

(a) 20 rad (b) 10 rad/s

☐

Question 5 12 pts

A scooter has wheels with a diameter of 120 mm. What is the angular speed of the wheels when the scooter is moving forward at 6.00 m/s?

hint: just the relationship linear speed and angular speed. 1 mm = 0.001m. you can check online how to convert revolution per minute to rad/s. 1 revolution is 2pi. 1 imin is 60s.

☐

100 rpm

☐

47.7 rpm

☐

50.0 rpm

☐

955 rpm

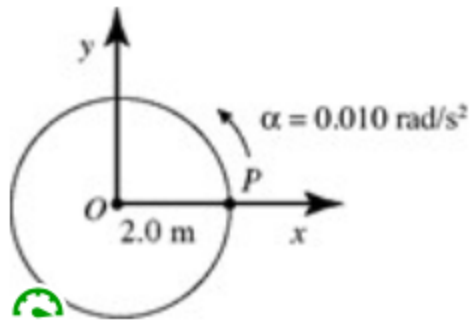
☐

72.0 rpm



Question 6 12 pts

In the figure, point P is at rest when it is on the x -axis. The linear speed of point P when it reaches the y -axis is closest to



hint: use kinematics. the angular displacement is $\pi/2$. To find the linear speed multiply the angular speed by R .



0.71 m/s.



0.24 m/s.



0.35 m/s.



0.49 m/s.

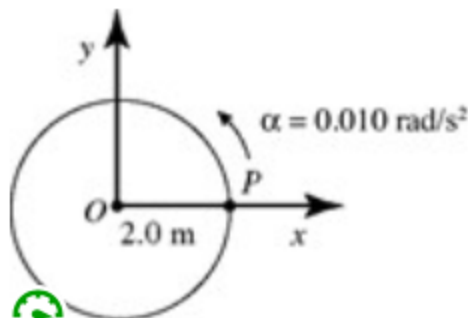


0.18 m/s.



Question 7 12 pts

In the figure, point P is at rest when it is on the x -axis. The time t , when P returns to the original position on the x -axis, is closest to



hint: use kinematics equation. The angular displacement is 2π

☐

50 s.

☐

18 s.

☐

25 s.

☐

35 s.

☐

13 s.



Question 8 12 pts

A 4.50-kg wheel that is 34.5 cm in diameter rotates through an angle of 13.8 rad as it slows down uniformly from 22.0 rad/s to 13.5 rad/s. What is the magnitude of the angular acceleration of the wheel?

☐

10.9 rad/s²

☐

0.616 rad/s²

☐

5.45 rad/s²

☐

111 rad/s²

☐

22.5 rad/s²



Question 9 12 pts

A bicycle has wheels that are 60 cm in diameter. What is the angular speed of these wheels when it is moving at 4.0 m/s?

☐

4.8 rad/s

☐

0.36 rad/s

☐

13 rad/s

☐

1.2 rad/s

☐

7.6 rad/s

Not saved

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